

Computational Fluid Dynamics for Aerospace Applications

Lecture 6 Stability

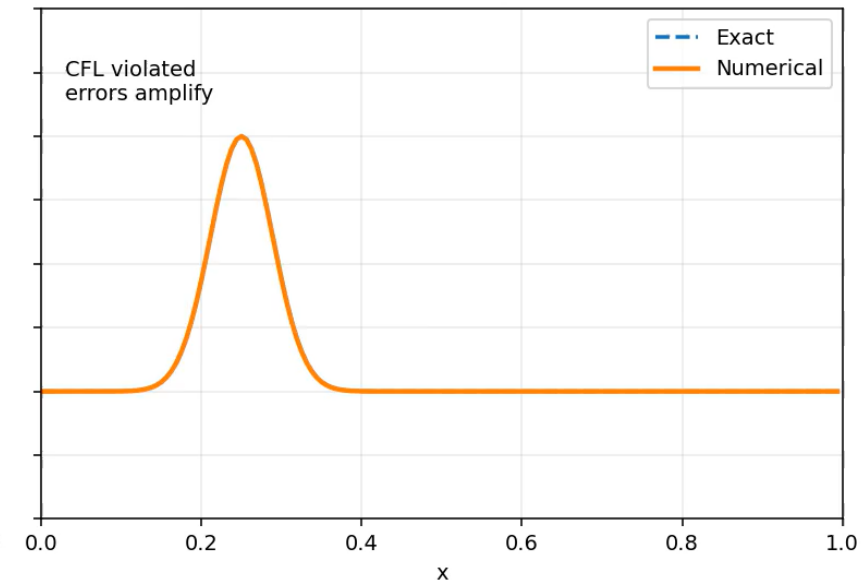
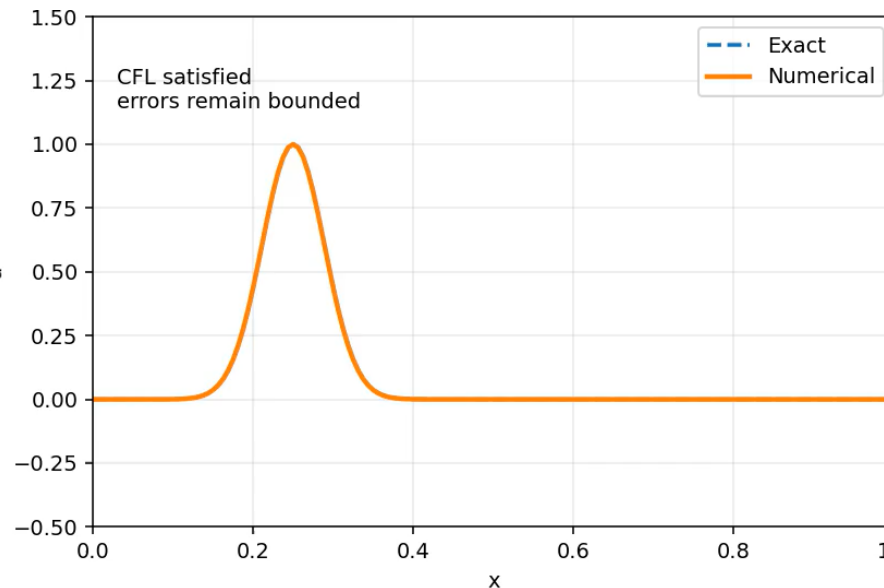
Stability

- Stability
- Explicit vs. Implicit

Stability

- We previously derived algebraic approximations to our PDEs and identified the rate at which their error shrinks as $\Delta x \rightarrow 0$
- Will these approximations work at any value of $\Delta t, \Delta x$? No!
- Not every approximation + step size is a good choice

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$



Stability

- **Stability**: A numerical scheme is stable if small errors introduced into the solution do not grow without bound as the computation advances
- If you're lucky, an unstable simulation will “blow-up” and crash
- If you're unlucky, it will run and return a bad answer
- How can we predict whether our approximation will be stable?
- Commonly achieved through von Neumann stability analysis

Stability

Note: $i \rightarrow \sqrt{-1}$

- Fourier Series tells us we can construct any variable (e.g., $u(x)$) as a combination of infinite sin/cos waves

$$u(x) = a_0 + \sum_{j=1}^{\infty} \left(a_j \cos\left(\frac{2\pi jx}{L}\right) + b_j \sin\left(\frac{2\pi jx}{L}\right) \right)$$



- But recall $e^{i\theta} = \cos\theta + i\sin\theta$ so we can equally write in one-dimension:

$$u(x) = \sum_{j=-\infty}^{\infty} c_j e^{i\frac{2\pi jx}{L}}$$

- Or two dimensions:

$$u(x, t) = \sum_{n=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} c_{j,n} e^{i\frac{2\pi jx}{L}} e^{i\frac{2\pi nt}{T}}$$

Stability

- One single wavenumber/frequency will then take the form:

indices $\leftarrow u_j^{(n)} = e^{\underline{i\alpha n \Delta t}} e^{ik \underline{j \Delta x}}$ exponents

$\downarrow$
 $u(j,n)$

$u_j^n = \underbrace{(e^{i\alpha \Delta t})^n}_{\text{constant } \sigma} \underbrace{(e^{ik \Delta x})^j}_{\text{constant } \beta}$

$u_j^n = \sigma^n \beta^j$

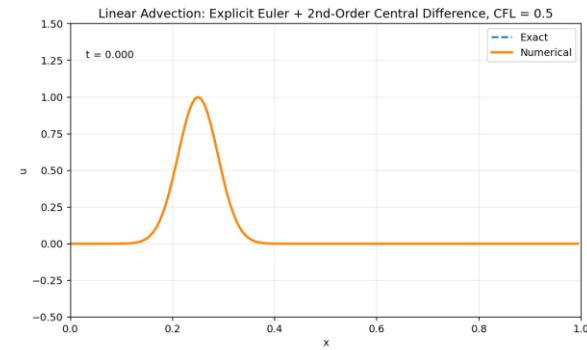
$$u_j^n = \sigma^n \beta^j$$

- Our solution is bounded only if $|\sigma| \leq 1$

Stability

$$\hat{u}_j^n = \sigma^n \beta^j$$

Unstable



- What is the stability of Explicit Euler + second order central difference applied to linear advection equation?

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

$$\frac{\hat{u}_j^{n+1} - \hat{u}_j^n}{\Delta t} + a \left(\frac{\hat{u}_{j+1}^n - \hat{u}_{j-1}^n}{2\Delta x} \right) = 0$$

$$\hat{u}_j^{n+1} = \hat{u}_j^n - \frac{a\Delta t}{2\Delta x} (\hat{u}_{j+1}^n - \hat{u}_{j-1}^n)$$

$$\sigma^{n+1} \beta^j = \sigma^n \beta^j - c [\sigma^n \beta^{j+1} - \sigma^n \beta^{j-1}]$$

$$(\sigma^n \beta^j) \sigma = \sigma^n \beta^j (1 - c [\beta - \beta^{-1}])$$

$$\sigma = 1 - c [\beta - \beta^{-1}]$$

$$\sigma = 1 - c [e^{ik\Delta x} - e^{-ik\Delta x}]$$

$$2i \sin(k\Delta x)$$

$$\sigma = 1 - \underbrace{2ic \sin k\Delta x}_{\text{imaginary}}$$

real

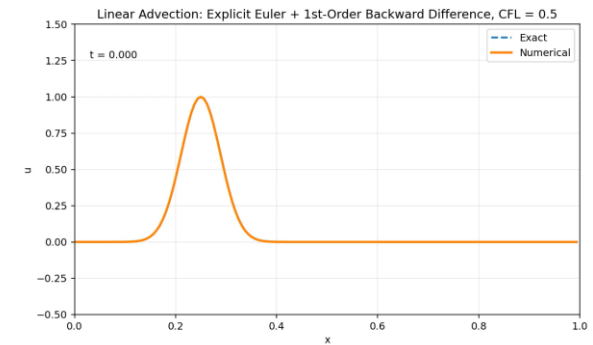
$$|\sigma| = \sqrt{(1)^2 + (c \sin(k\Delta x))^2}$$

$$|\sigma| > 1$$

Stability

$$\hat{u}_j = \sigma^n \beta^j$$

$$C = \frac{a \Delta t}{\Delta x}$$



- What is the stability of Explicit Euler + first order backward difference applied to linear advection equation?

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

$$\frac{u_j^{n+1} - \hat{u}_j^n}{\Delta t} + a \left(\frac{\hat{u}_j^n - \hat{u}_{j-1}^n}{\Delta x} \right) = 0$$

$$u_j^{n+1} = \hat{u}_j^n - \boxed{\frac{a \Delta t}{\Delta x}} (\hat{u}_j^n - \hat{u}_{j-1}^n)$$

$$u_j^{n+1} = \hat{u}_j^n - C [\hat{u}_j^n - \hat{u}_{j-1}^n]$$

$$u_j^{n+1} = \hat{u}_j^n (1 - C) + C \hat{u}_{j-1}^n$$

$$\sigma^{n+1} \beta^j = \sigma^n \beta^j (1 - C) + C \sigma^n \beta^{j-1}$$

$$(\sigma^n \beta^j) \sigma = (\sigma^n \beta^j) [(1 - C) + C \beta^{-1}]$$

$$\sigma = 1 - C + C \beta^{-1}$$

$$\sigma = 1 - C + C (e^{-ik\Delta x})$$

$$\sigma = 1 - C + C [\cos(k\Delta x) - i \sin(k\Delta x)]$$

$$|\sigma| = \sqrt{(1 - C + C \cos(k\Delta x))^2 + (C \sin(k\Delta x))^2}$$

$$|\sigma| = \sqrt{(1 - C)^2 + 2C(1 - C)\cos(k\Delta x) + C^2(\cos^2(k\Delta x) + \sin^2(k\Delta x))}$$

Algebra

$$|\sigma| = \sqrt{1 - 2C(1 - C)(1 - \cos(k\Delta x))}$$

Need to be ≥ 0 ≥ 0

$$2C(1 - C) \geq 0 \Rightarrow 0 \leq C \leq 1$$

$$\Delta t, \Delta x > 0$$

Stability

- Explicit Euler, first order backward difference stable for:

$$0 \leq c = \frac{a\Delta t}{\Delta x} \leq 1$$

unstable for $a < 0$

stable for $a \geq 0$ if $0 \leq c \leq 1$

- If we repeated for Explicit Euler, first order forward difference:

$$-1 \leq c = \frac{a\Delta t}{\Delta x} \leq 0$$

stable for $a \leq 0$ if $-1 \leq c \leq 0$

unstable for $a > 0$

Stability

- We just showed the linear advection equation solved with explicit euler could be stable or unstable depending on whether we use a forward or backward spatial discretization
- For advection, we often use a scheme that locally switches between forward and backward depending on the local velocity
- This is called an upwind scheme
 - If $a > 0$, use a backward discretization
 - If $a < 0$, use a forward discretization

Stability

$$\frac{a\Delta t}{\Delta x} \leq 1$$

- The parameter on the left is extremely important and variations of it will show up frequently in CFD
- Generally called the Courant-Friedrichs-Lewy (CFL) condition

$$CFL = \frac{|u|\Delta t}{\Delta x}$$

- Note u vs. a , this should be rate of propagation of information
- Note that every combination of a temporal/spatial scheme will lead to a different stability condition (i.e., not always $CFL < 1$)

Stability

Note: It's common to hear that stability requires $CFL < 1$ for CFD. This is not always true! The stable values of CFL depend on the exact combination of temporal and spatial approximation

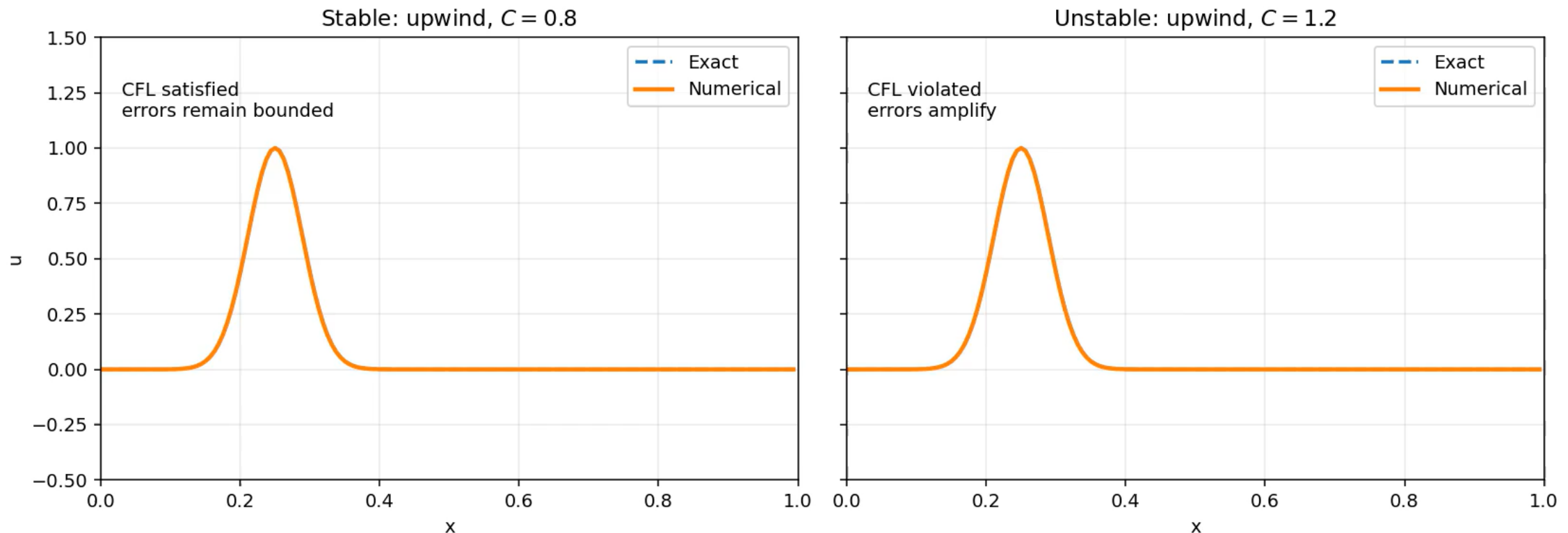
- The CFL number is generally used to decide the timestep size
- For example:

$$\text{if } \frac{a\Delta t}{\Delta x} \leq 1, \text{ then } \Delta t \leq \frac{\Delta x}{a}$$

- We pick a grid (Δx) to resolve some scale/physics of interest
- If a is a known constant, easy
- If a is unknown/varies, compute $\Delta x/a$ at every point at each time and pick the most restrictive result (i.e. $\min(\Delta t)$)
- In practice, go lower than required to be safe (e.g., $CFL = 0.8$)
- For arbitrary CFL number, $\Delta t = CFL \times \frac{\Delta x}{a}$

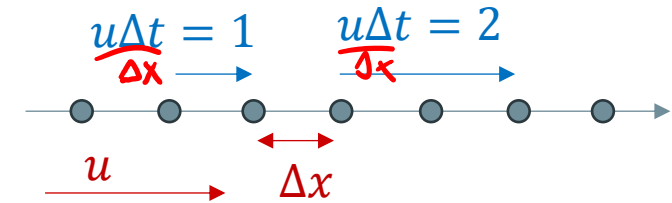
Stability

- Linear advection, explicit euler, first order upwind ($CFL \leq 1$)



Stability

$$CFL = \frac{u\Delta t}{\Delta x}$$

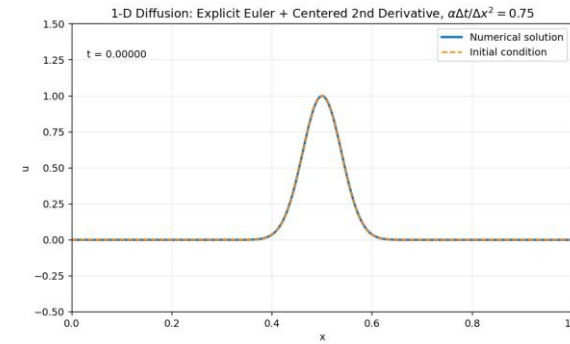
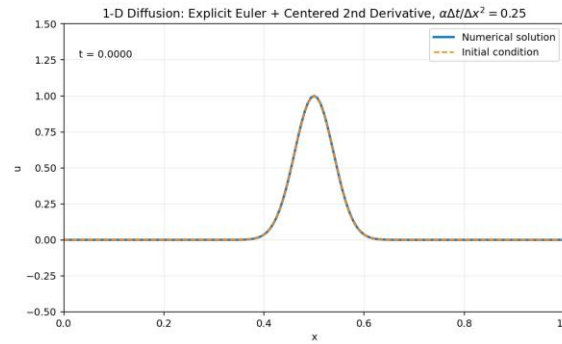


- What does the CFL number intuitively represent?
- $u\Delta t$ represents the distance information travels over a time Δt
- The ratio of $u\Delta t$ and Δx represents how many gridpoints information travels over a time Δt
- How many gridpoints should information travel each Δt ?
- Backwards difference: $u_j^{n+1} = u_j^n - \frac{a\Delta t}{\Delta x} (u_j^n - u_{j-1}^n)$
- This scheme only knows about information *one* gridpoint away (i.e., between j and $j - 1$), so propagating beyond that would lead to a loss of information (and instability)

Stability

$$u_j^n = \hat{\sigma} \hat{\beta}^j$$

$$\beta = e^{ik\Delta x}$$



- What is the stability of Explicit Euler + first order central difference applied to the unsteady diffusion equation?

$$\frac{\partial u}{\partial t} = \Gamma \frac{\partial^2 u}{\partial x^2}$$

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \Gamma \left(\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2} \right)$$

$$u_j^{n+1} = u_j^n + \left[\frac{\Gamma \Delta t}{\Delta x^2} \right] (u_{j+1}^n - 2u_j^n + u_{j-1}^n)$$

$$\hat{\sigma}^{n+1} \hat{\beta}^j = \hat{\sigma}^n \hat{\beta}^j + C [\hat{\sigma}^n \hat{\beta}^{j+1} - 2\hat{\sigma}^n \hat{\beta}^j + \hat{\sigma}^n \hat{\beta}^{j-1}]$$

$$\sigma = 1 + C [\underbrace{\beta + \beta^{-1}}_{2\cos(k\Delta x)} - 2]$$

$$\sigma = 1 + 2C \left[\cos(k\Delta x) - 1 \right]$$

Another trig. identity $= -2 \sin^2 \left(\frac{k\Delta x}{2} \right)$

$$\sigma = 1 - 4C \sin^2 \left(\frac{k\Delta x}{2} \right)$$

0 to 1

works
case = 1

$$|\sigma| \leq 1 \rightarrow -1 \leq \sigma \leq 1$$

$$\rightarrow 0 \leq 4C \sin^2 \left(\frac{k\Delta x}{2} \right) \leq 2$$

$$\downarrow$$

$$0 \leq C \leq \frac{1}{2}$$

Stable for $\frac{\Gamma \Delta t}{\Delta x^2} \leq \frac{1}{2}$

Stability

↓ diffusive
viscous CFL #
↓

$$\frac{\Gamma \Delta t}{\Delta x^2} \leq \frac{1}{2}$$

- The term on the left is often informally called the “diffusive CFL”
- We can figure out the required timestep using this condition

$$\Delta t \leq \frac{\Delta x^2}{2\Gamma}$$

- If Γ or Δx vary across the domain, calculate Δt everywhere and use the minimum value
- For arbitrary CFL condition, $\Delta t = CFL \times \frac{\Delta x^2}{\Gamma}$

Stability

- What about the real Navier-Stokes equations?
 - Many more terms of higher complexity
 - Multiple simultaneous equations to solve
- In practice, we usually define stability for each term individually, calculate them all, and pick the most restrictive
 - Calculate advective CFL numbers for x , y , and z
 - Calculate diffusive CFL numbers for x , y , and z
 - Ignore other terms
- In theory, this misses important interactions between the terms
- In practice, add larger factors of safety to ensure stability

Stability

- Recall that in Navier-Stokes, information propagates at the speed of sound + advection
- The correct advective CFL number can be derived as:

$$CFL = \frac{(|u| + a)\Delta x}{\Delta t}$$

- For Navier-Stokes, u will be different at every point/time
 - Need to calculate CFL everywhere and pick the most restrictive Δt

Stability

- In practice, we generally solve the following equations and pick the single most restrictive to use as our timestep (with a FOS)

$$\Delta t_{\text{conv}} = CFL \times \min \left(\frac{\Delta x}{|u| + a}, \frac{\Delta y}{|v| + a}, \frac{\Delta z}{|w| + a} \right)$$

$$\Delta t_{\text{diff}} = CFL \times \frac{1}{\max(\nu, k)} \times \min(\Delta x^2, \Delta y^2, \Delta z^2)$$

$$\Delta t = \min(\Delta t_{\text{conv}}, \Delta t_{\text{diff}})$$

Explicit vs. Implicit

- Explicit: A discretization scheme that computes the next time solution from known quantities (everything we've done so far)
 - For example: $u_j^{n+1} = u_j^n - \frac{a\Delta t}{\Delta x} (u_j^n - u_{j-1}^n)$
 - We solve for one variable representing the next time (LHS, unknown) using terms from the previous time (RHS, all known variables)
- Implicit: A discretization scheme that computes the next time solution using unknown quantities
 - For example, linear advection, backward Euler & SO central difference
 - $\frac{u_j^{n+1} - u_j^n}{\Delta t} + a \frac{u_{j+1}^{n+1} - u_{j-1}^{n+1}}{2\Delta x} = 0$
 - The quantity we want (u_j^{n+1}) is a function of unknowns ($u_{j-1}^{n+1}, u_{j+1}^{n+1}$)

Explicit vs. Implicit

For simplicity, solvers that we write ourselves will only be explicit in this course

- Such schemes can still be solved, but implicitly
 - Simultaneous equations for every point solved together each timestep
 - This is computationally more complex, but many packages exist that will solve the system for you
- Why bother? Implicit schemes often have better stability criteria
 - Linear advection – Explicit Euler, SO central: Always unstable
 - Linear advection – Backward Euler, SO central: Always stable
- Large timesteps are a mixed bag
 - Can simulate more time (e.g., several engine cycles instead of one)
 - Error is still proportional to Δt
 - Fast physics is missed if the timestep is too large

Looking Forward

- Even a stable solution can still be distorted in two ways:
 - Change in amplitude
 - Change in wavespeed
- We will introduce dissipation and dispersion, which govern this
- We still haven't figured out what to do at boundaries
- Once we do the above, you have the tools to write a CFD code